Zuhao Yang (杨祖豪)

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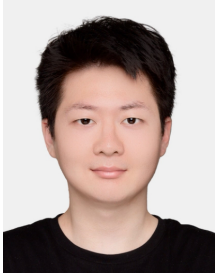

Educational Background

Nanyang Technological University 2024.01 – 2028.01
Doctor of Philosophy in Computer Science and Engineering Supervisor: Prof. Shijian Lu
- Video-Centric Research: temporal grounding, complex reasoning, agentic tool use, and generation.

Nanyang Technological University 2022.08 – 2024.01
Master of Science in Artificial Intelligence cGPA: 4.8/5.0

University of Alberta 2017.09 – 2021.06
Bachelor of Science with Specialization in Computing Science cGPA: 3.7/4.0
- Outstanding Graduate with Distinction, Dean's Honor Roll, and International Student Scholarship.

Selected Publications

Large Multimodal Models: Video Understanding, Reasoning, Agents, and Efficiency

- *LongVT: Incentivizing "Thinking with Long Videos" via Native Tool Calling*.
  Zuhao Yang*, Sudong Wang*, Kaichen Zhang*, ..., Xingxuan Li, Lidong Bing. **CVPR 2026**
- *OpenMMReasoner: Pushing the Frontiers for Multimodal Reasoning with an Open and General Recipe*.
  Kaichen Zhang*, Keming Wu*, **Zuhao Yang**, ..., Xingxuan Li, Lidong Bing. **CVPR 2026**
- *SVAgent: Storyline-guided Long Video Understanding via Cross-modal Multi-agent Collaboration*.
  Zhongyu Yang, **Zuhao Yang**, Shuo Zhan, Tan Yue, Wei Pang, Yingfang Yuan. **CVPR 2026**
- *A Comprehensive Study on Visual Token Redundancy for Discrete Diffusion-based MLLMs*.
  Duo Li*, **Zuhao Yang***, Xiaoqin Zhang, Ling Shao, Shijian Lu. **CVPR 2026**
- *ToDRE: Effective Visual Token Pruning via Token Diversity and Task Relevance*.
  Duo Li*, **Zuhao Yang***, Xiaoqin Zhang, Ling Shao, Shijian Lu. **arXiv 2025**
- *TimeExpert: An Expert-Guided Video LLM for Video Temporal Grounding*.
  Zuhao Yang, Yingchen Yu, Yunqing Zhao, Shijian Lu, Song Bai. **ICCV 2025**

Deep Generative Models: Image, Video, and 3D Generation

- *Versatile Transition Generation with Image-to-Video Diffusion*.
  Zuhao Yang, Jiahui Zhang, Yingchen Yu, Shijian Lu, Song Bai. **ICCV 2025**
- *ReChar: Revitalising Characters with Structure-Preserved Aesthetic Enhancements*.
  Zhongyu Yang, Junhao Song, Zhang Luo, **Zuhao Yang**, et al. **SIGGRAPH Asia 2025**
- *AI-Generated Images as Data Sources: The Dawn of Synthetic Era*.
  Zuhao Yang, ..., Xiaoqin Zhang, Ling Shao, Shijian Lu. **TPAMI (2/3 major revision)**
- *PaCaNet: A Study on CycleGAN with Transfer Learning for Diversifying Fused Chinese Painting*.
  Zuhao Yang*, Huajun Bai*, Zhang Luo, Yang Xu, Wei Pang, ..., Yingfang Yuan. **arXiv 2023**

Natural Language Evaluation: Open-Ended Generation and Question-Answering Tasks

- *Evaluating Text Generation Quality Using Spectral Distances of Surprisal*.
  Zhichen Liu, Yongyuan Li, Yang Xu, Yu Wang, Yingfang Yuan, **Zuhao Yang**. **EMNLP 2025**
- *QAEval: Mixture of Evaluators for Question-Answering Task Evaluation*.
  Tan Yue, Rui Mao, Xuzhao Shi, Shuo Zhan, **Zuhao Yang**, Dongyan Zhao. **ACL 2025**
- *FACE: Evaluating Natural Language Generation with Fourier Analysis of Cross-Entropy*.
  Zuhao Yang*, Yingfang Yuan*, Yang Xu*, Shuo Zhan, Huajun Bai, Kefan Chen. **NeurIPS 2023**

Work Experience

Shanda AI Research Institute & MiroMind 2025.04 – 2026.02
AI Scientist Intern
- Researched multimodal reasoning and agentic tool use for large multimodal models.
- Collaborated with Dr. Lidong Bing and contributed to MiroThinker (RL infrastructure) development.

ByteDance Inc. & TikTok 2023.11 – 2025.03

Computer Vision Research Intern

- Researched video generation with diffusion models and video temporal grounding with MoEs.
- Collaborated with Dr. Song Bai and filed two patents (one US, one SG) upon project completion.

TMI Robotics Technology Co. Ltd 2021.05 – 2022.06

Natural Language Processing Research Engineer

- Developed the dialogue system of a medical robot (FAQ Service: Elastic Search + Sentence-BERT; Guide Service: TextRCNN + Knowledge Graph) and built department-specific knowledge graphs via named entity recognition, relation extraction and entity clustering. Filed one CN patent.
- Optimized model inference (OpenVINO, TensorRT) and dialogue pipeline (Rasa) with BERT compression and weight pruning.

Additional Information

Open-Source Projects

- **Core Contributor**, LMMs-Eval **(3.7k+ ★)**: a unified evaluation toolkit for multimodal models across text, image, video, and audio tasks.
- **Core Contributor**, LMMs-Engine **(700+ ★)**: a lean and flexible training engine for large multimodal models, built for rapid prototyping at scale.

Academic Services

- **Reviewer**: CVPR 24/25/26, ECCV 24/26, ACM MM 24/25, NeurIPS 24/25, ICLR 25, AISTATS 25/26, ICML 25, ICCV 25, BMVC 26; IEEE TPAMI, Pattern Recognition, SPIE JEI.
- **PC Member**: SyntaGen, Neural Rendering Intelligence, Engineering Agentic Intelligence.

Filed Patents

- *Method, Device, and Medium for Video Temporal Grounding with Mixture of Experts.* US Patent.
- *Method, Device, and Medium for Generating Transition Videos with Diffusion Model.* SG Patent.
- *Method, Device, and Medium for Automatic Question-Answering.* CN Patent.